Level 1 – Task 1: Top Cuisines

Objective:

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Find the top 3 most common cuisines and calculate the percentages of restaurants serving each of those cuisines.

Python code: Include both code blocks one after the other

Python Code 1

```
top_cuisines=df['Cuisines'].value_counts().head(3)

print(top_cuisines)
```

Python Code 2

```
total_restaurants = len(df)

top_cuisine_percentage = (df['Cuisines'].value_counts().head(3) / total_restaurants) * 100

print(top_cuisine_percentage)
```

Output 1

The top 3 cuisines with their counts.

```
Cuisines
North Indian              936
North Indian, Chinese     511
Chinese                   354
Name: count, dtype: int64
```

Output 2

The percentages of those top 3 cuisines.

```
Cuisines

North Indian              9.800021

North Indian, Chinese     5.350225

Chinese                   3.706418

Name: count, dtype: float64
```

Conclusion:

The analysis shows that the top three most common cuisines in the dataset are North Indian, North Indian, Chinese, and Chinese. Among these, North Indian is the most popular cuisine, with 936 restaurants (approximately 9.80%) serving it. This indicates that North Indian cuisine has the highest representation in the dataset, followed by North Indian–Chinese and Chinese cuisines.

Task 2: City Analysis

Objective

Find the city with the highest number of restaurants, calculate the average restaurant rating for each city, and identify the city with the highest average rating.

Python Code 1

```
city_count = df['City'].value_counts()

print(city_count.head())
```

Output 1

```
City

New Delhi      5473

Gurgaon        1118

Noida          1080

Faridabad       251

Ghaziabad        25

Name: count, dtype: int64
```

Python Code 2

```
average_rating = df.groupby('City')['Aggregate
rating'].mean().sort_values(ascending=False)print(average_rating.head())
```

Output 2

```
City
Inner City              4.900000
Quezon City             4.800000
Makati City             4.650000
Pasig City              4.633333
Mandaluyong City        4.625000
Name: Aggregate rating, dtype: float64
```

Python Code 3

```
highest_avg_rating = average_rating.idxmax()
print("City with the highest average rating:", highest_avg_rating)
```

Output 3

City with the highest average rating: Inner City

Conclusion

The analysis shows that New Delhi has the highest number of restaurants in the dataset. The average restaurant rating was calculated for each city, and Inner City was identified as the city with the highest average restaurant rating based on the dataset.

Task 3: Price Range Distribution

Objective

To analyze the distribution of restaurants across different price range categories by calculating the percentage of restaurants in each category and visualizing the distribution using a bar chart.

Python Code 1

```
# Count restaurants in each price range

price_count = df['Price range'].value_counts()


# Calculate percentage

price_percentage = (price_count / len(df)) * 100


print(price_percentage)


# Bar chart

import matplotlib.pyplot as plt


plt.bar(price_percentage.index, price_percentage.values)


plt.xlabel("Price Range")
```

```
plt.ylabel("Percentage of Restaurants")

plt.title("Price Range Distribution")


plt.savefig('price_range_distribution.png', dpi=300, bbox_inches='tight')

plt.show()
```

Output

(percentage table)

Price range

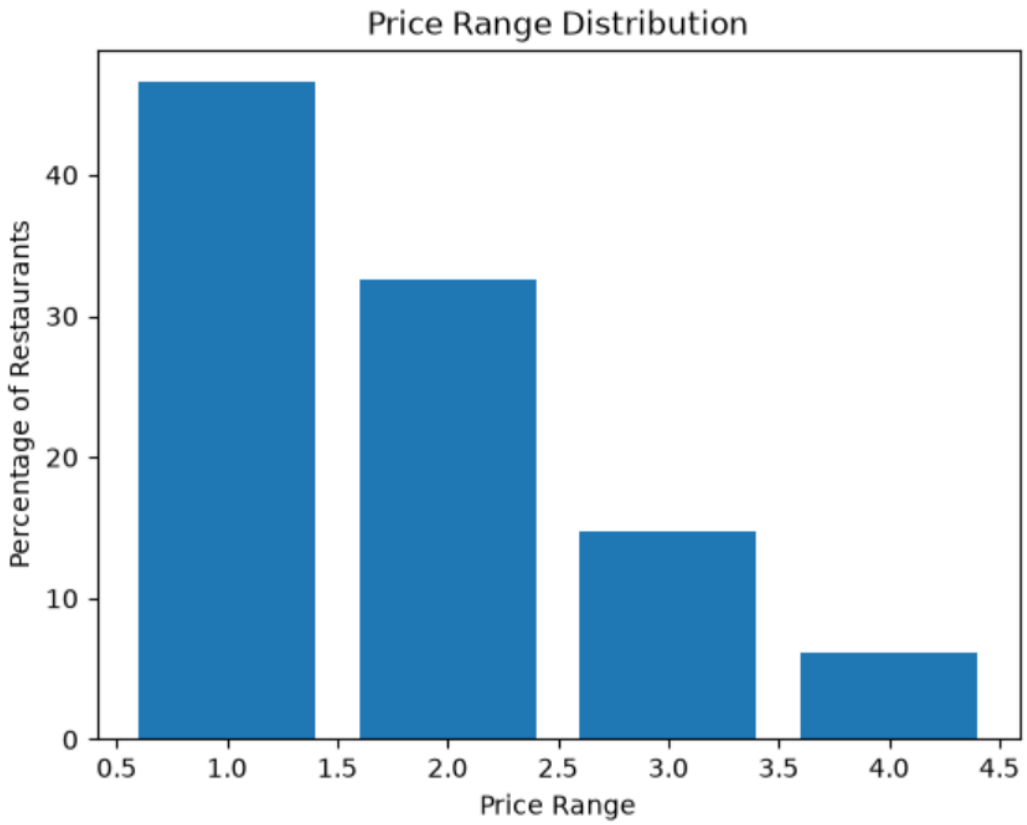
1 46.529159

2 32.593446

3 14.741912

4 6.135483

Name: count, dtype: float64



Conclusion

The distribution of restaurants across different price ranges was analyzed. The percentage of restaurants in each price range was calculated and visualized using a bar chart, making it easy to compare the distribution.

Task 4: Online Delivery

Objective

To determine the percentage of restaurants that offer online delivery and compare the average ratings of restaurants with and without online delivery.

Python Code 1

```
import pandas as pd

df = pd.read_csv("Dataset .csv")

# Percentage of restaurants offering online delivery

online_delivery = (df['Has Online delivery'].value_counts(normalize=True) * 100).round(2)

print(online_delivery)
```

Output 1

```
Has Online delivery
No     74.34
Yes    25.66
Name: proportion, dtype: float64
```

Python Code 2

```
# Average rating based on online delivery


avg_rating = df.groupby('Has Online delivery')['Aggregate rating'].mean().round(2)


print(avg_rating)
```

Output 2

```
Has Online delivery
No    2.47
Yes   3.25
Name: Aggregate rating, dtype: float64
```

Conclusion

The percentage of restaurants providing online delivery was calculated. The average ratings of restaurants with and without online delivery were compared to understand whether online delivery has any relationship with customer ratings.

Level 2 – Task 1: Restaurant Ratings

Objective

To analyze the distribution of restaurant ratings and calculate the average number of votes received by restaurants.

Python Code 1

```
# Distribution of Aggregate Ratings

rating_distribution = df['Aggregate rating'].value_counts().sort_index()

print(rating_distribution)
```

Output 1

```
Aggregate rating
0.0    2148
1.8       1
1.9       2
2.0       7
2.1      15
2.2      27
2.3      47
2.4      87
2.5     110
2.6     191
2.7     250
```

```
2.8    315

2.9    381
3.0    468
3.1    519
3.2    522
3.3    483
3.4    498
3.5    480
3.6    458
3.7    427
3.8    400
3.9    335
4.0    266
4.1    274
4.2    221
4.3    174
4.4    144
4.5     95
4.6     78
4.7     42
4.8     25
4.9     61
Name: count, dtype: int64
```

Python Code 2

```
# Average number of votes received


average_votes = df['Votes'].mean()


print("Average number of votes:", round(average_votes, 2))
```

Output 2

Average number of votes: 156.91

Conclusion

The distribution of aggregate ratings was analyzed, and the average number of votes received by restaurants was calculated. This helps in understanding customer ratings and restaurant popularity.

Task 2: Cuisine Combination

Objective

To identify the most common combinations of cuisines served by restaurants and analyze their frequency.

Python Code

```
# Find the most common cuisine combinations

cuisine_combination = df['Cuisines'].value_counts().head(10)

print(cuisine_combination)
```

Output

```
Cuisines
North Indian                936
North Indian, Chinese       511
Chinese                     354
Fast Food                   354
```

North Indian, Mughlai	334
Cafe	299
Bakery	218
North Indian, Mughlai, Chinese	197
Bakery, Desserts	170
Street Food	149

Name: count, dtype: int64

Conclusion

The most common cuisine combinations served by restaurants were identified by analyzing the cuisine data. This analysis helps understand the popular cuisine combinations and the food preferences offered by restaurants.

Task 3: Geographic Analysis

Objective

To plot the locations of restaurants on a map using latitude and longitude coordinates and identify any patterns or clusters of restaurants in specific areas.

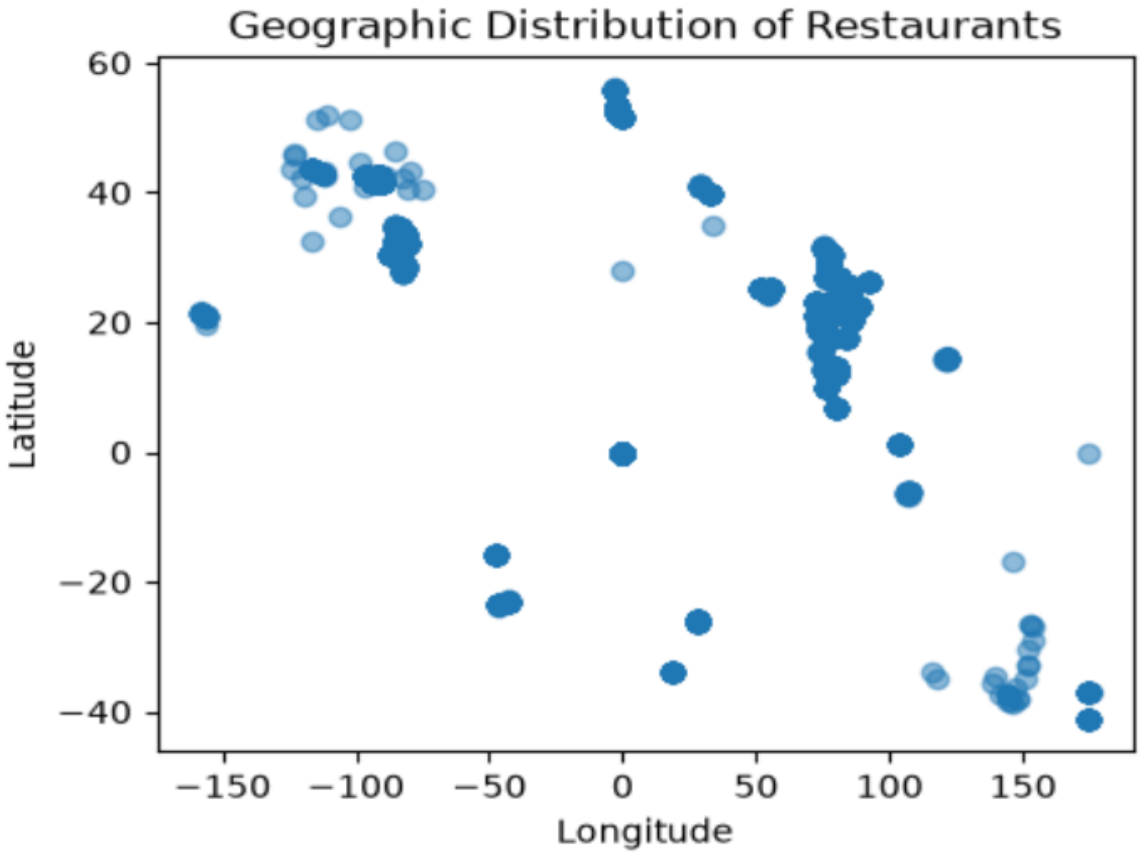
Python Code

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))
plt.scatter(df['Longitude'], df['Latitude'], alpha=0.5)

plt.title("Geographic Distribution of Restaurants")
plt.xlabel("Longitude")
plt.ylabel("Latitude")

plt.show()
```



Conclusion

The restaurant locations were visualized using latitude and longitude coordinates. The plot helps identify areas where restaurants are concentrated and shows the geographic distribution of restaurants across different locations.

Task 4:Restaurant Chains

Objective

To identify the restaurant chains present in the dataset and analyze their average ratings and popularity.

Python Code 1

```
# Find restaurant chains


chains = df['Restaurant Name'].value_counts()


print(chains.head(10))
```

Output 1

```
Restaurant Name
Cafe Coffee Day         83
Domino's Pizza          79
Subway                  63
Green Chick Chop        51
```

McDonald's 48

Keventers 34

Pizza Hut 30

Giani 29

Baskin Robbins 28

Barbeque Nation 26

Name: count, dtype: int64

Python Code 2

```
# Average rating of top restaurant chains

top_chains = df['Restaurant Name'].value_counts().head(10).index

ratings = df[df['Restaurant Name'].isin(top_chains)]

print(ratings.groupby('Restaurant Name')['Aggregate rating'].mean())
```

Output 1

Restaurant Name

Barbeque Nation 4.353846

Baskin Robbins 1.860714

Cafe Coffee Day 2.419277

Domino's Pizza 2.740506

Giani 2.689655

Green Chick Chop 2.672549

Keventers 2.870588

McDonald's 3.339583

Pizza Hut 3.320000

Subway 2.907937

Name: Aggregate rating, dtype: float64

Conclusion

The restaurant chains present in the dataset were identified, and their average ratings were analyzed. This analysis helps understand the popularity and performance of different restaurant chains.